

Cyclistic Bike-Share Case Study

How Annual Members and Casual Riders Use Cyclistic Bikes Differently

Tools: BigQuery (SQL) • Google Sheets

1. Executive Summary

Cyclistic wants to understand how annual members and casual riders use its bike-share service differently so that the company can develop strategies to encourage casual riders to become annual members.

I analyzed **over 6 million bike-share trips** from August 2025 through July 2026 using BigQuery and SQL. The analysis examined ride volume, ride duration, day-of-week and time-of-day patterns, monthly trends, bike preferences, and start-station usage.

The analysis found that annual members account for **64.38% of rides**, compared with **35.62% for casual riders**, and show more regular weekday and time-of-day usage patterns. Casual riders typically take longer rides, use the service more heavily toward the weekend, show greater seasonal variation, and have different high-volume start locations. Both groups predominantly use electric bikes, making bike preference a smaller differentiator.

Based on these findings, Cyclistic should test targeted membership promotions during periods of high casual ridership, prioritize high-volume casual start stations for marketing, and tailor membership messaging toward casual riders taking longer-duration trips.

2. Business Task

Cyclistic's marketing strategy is focused on increasing the number of annual members. To support this goal, the analysis addresses the following business question:

How do annual members and casual riders use Cyclistic bikes differently?

The objective is to identify meaningful differences in riding behavior that could help Cyclistic develop data-informed marketing strategies for converting casual riders into annual members.

3. Data and Tools

The analysis used 12 months of historical bike-share trip data covering **August 2025 through July 2026**. The dataset is based on publicly available Divvy trip data provided for the Cyclistic case study and contains information including ride IDs, bike types, start and end timestamps, station information, geographic coordinates, and rider type.

The 12 monthly datasets originally contained **6,037,968 trip records**.

Tools used:

- **Google BigQuery / SQL** — data validation, cleaning, transformation, and analysis
- **Google Sheets** — creation of presentation-ready visualizations
- **Google Drive / Google Docs** — project organization and documentation

The dataset does not contain personally identifiable rider information. Therefore, the analysis can compare trip behavior between rider categories but cannot identify individual riders or determine how frequently a specific person uses the service.

4. Data Preparation and Cleaning

The 12 monthly datasets were imported into BigQuery and validated before analysis. I first confirmed that all monthly tables shared a consistent 13-column schema, then combined them into a single dataset using SQL.

Data-quality checks identified **35 exact duplicate records** and **1,930 rides with non-positive durations**. These records were excluded from the cleaned dataset. Missing station information was retained because those rides remained valid for analyses that did not require station data; NULL station names were excluded only from station-specific analyses.

I created additional fields for ride duration in seconds and minutes and for day of week. After cleaning, the dataset contained **6,036,003 unique rides**, with no duplicate ride IDs and no non-positive ride durations.

A subsequent date-range validation identified **128 valid rides that started on July 31, 2025 and ended on August 1, 2025**. These records were retained in the cleaned dataset, but final analyses were restricted to rides starting from August 1, 2025 through July 31, 2026. This produced **6,035,875 rides** within the final analysis period.

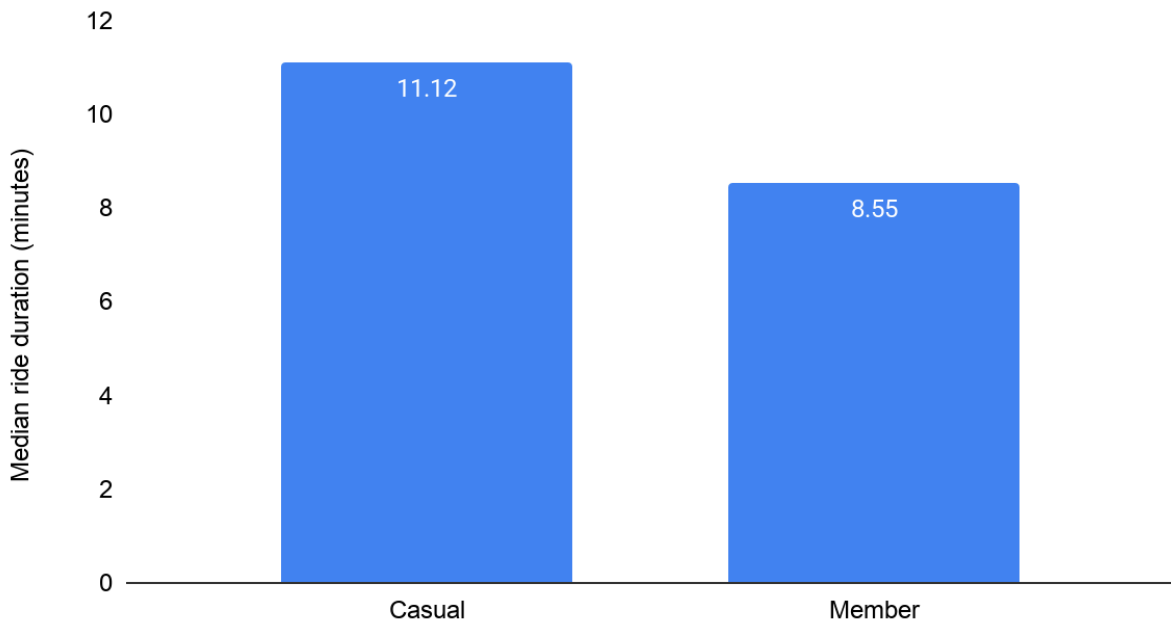
5. Analysis and Key Findings

5.1 Casual Riders Typically Take Longer Rides

Casual riders had a median ride duration of **11.12 minutes**, compared with **8.55 minutes** for annual members. This means the typical casual ride was approximately **30% longer** than the typical member ride.

Median duration was used because it is less affected by unusually long rides than the mean. The result indicates a clear difference in ride duration between the two groups, although duration alone does not reveal the purpose of a trip.

Casual riders typically take longer rides than members

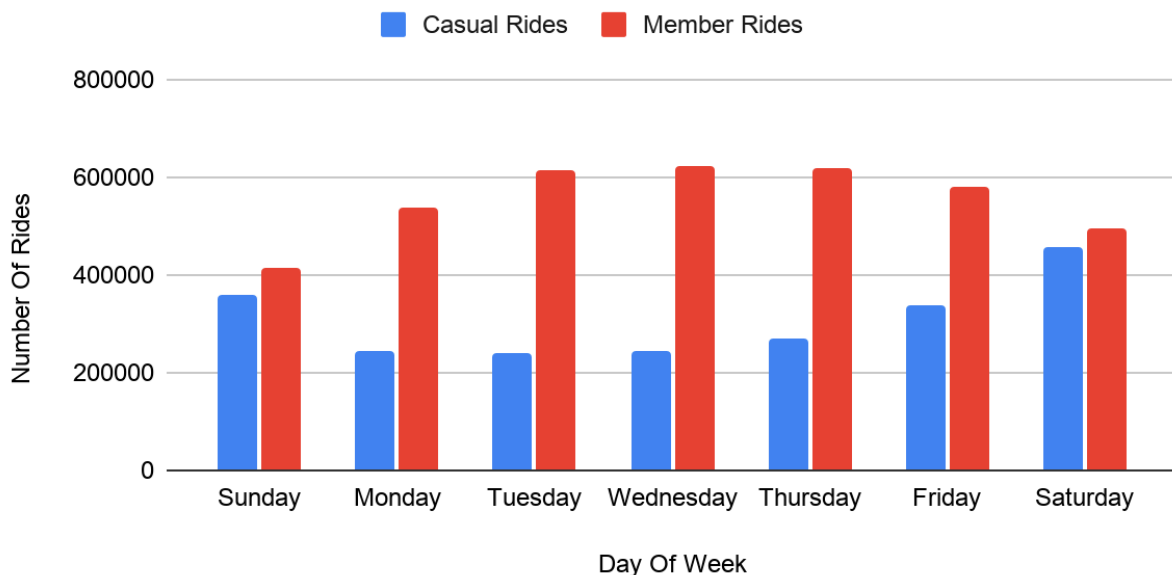


5.2 Members Are Stronger on Weekdays, While Casual Usage Rises Toward the Weekend

Ride patterns differed noticeably across the week. Annual-member activity was strongest during weekdays, with **Wednesday recording the highest member ride volume**. Casual ridership increased toward the weekend, with **Saturday recording the highest casual ride volume**.

This indicates that members and casual riders have different weekly usage patterns. Members show greater weekday activity, while casual usage is more concentrated toward the weekend. The data does not contain trip-purpose information, so these patterns should not be interpreted as direct evidence of commuting or recreational travel.

Member rides are strongest on weekdays, while casual rides rise toward the weekend



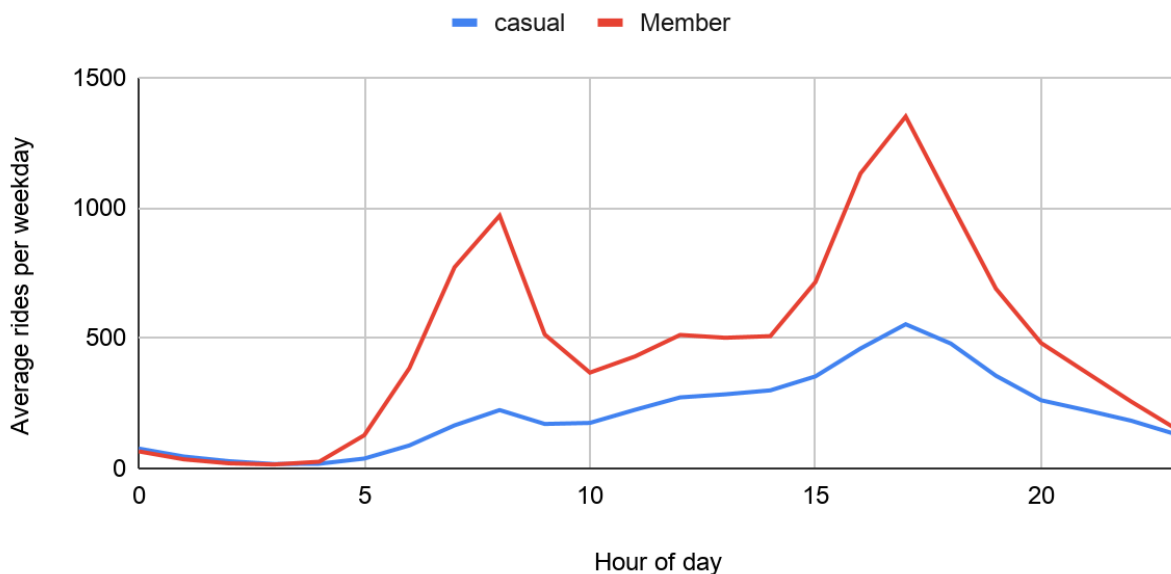
5.3 Member Weekday Usage Shows Distinct Time-of-Day Peaks

Looking specifically at weekdays revealed a stronger difference in time-of-day behavior. Annual members showed pronounced peaks in both the morning and late afternoon. Member usage averaged approximately **969.4 rides per weekday at 8:00 AM** and reached approximately **1,349.8 rides per weekday at 5:00 PM**.

Casual weekday usage increased more gradually throughout the day, reaching its highest level at approximately **552.9 rides per weekday at 5:00 PM**. These patterns suggest that member usage is more strongly structured around particular weekday times, while casual usage is distributed more gradually throughout the day.

The morning and late-afternoon member peaks are consistent with routine weekday travel patterns, but the dataset does not contain trip-purpose information and therefore cannot confirm that these rides represent commuting.

Member weekday rides show distinct morning and afternoon peaks



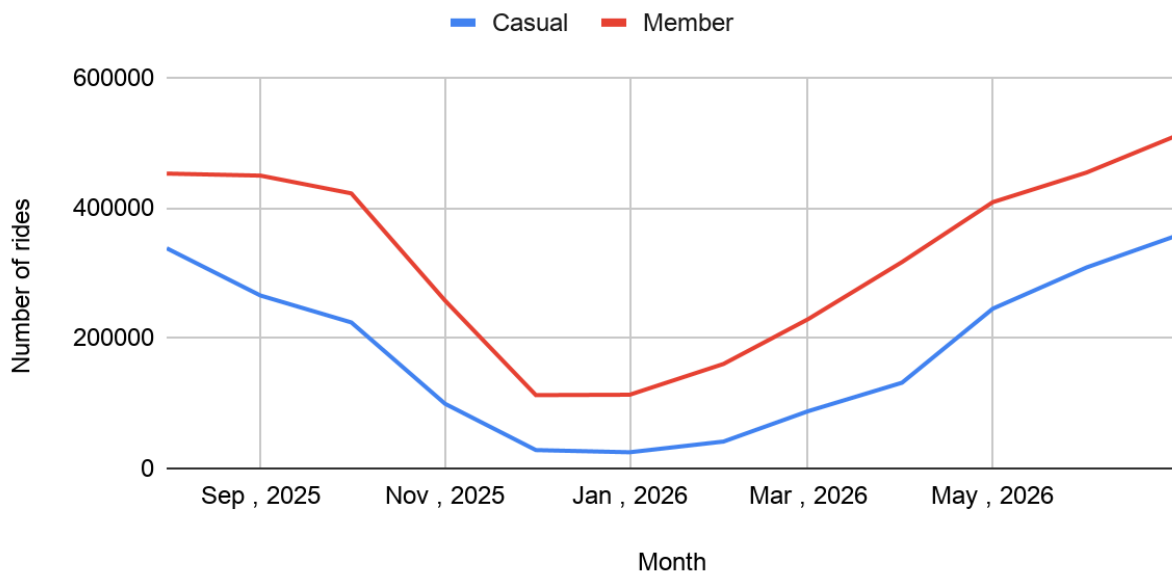
5.4 Casual Ridership Shows Greater Monthly Variation

Both rider groups showed substantial changes in ride volume throughout the year. Ridership declined during the later months of 2025, reached its lowest levels around December and January, and then increased through the first half of 2026. **July 2026 recorded the highest monthly ride volume for both groups, with 357,534 casual rides and 511,041 member rides.**

Casual ridership showed greater variation across the year than member ridership. **The pattern is consistent with stronger seasonality among casual riders**, while member usage remains comparatively more consistent throughout the year.

Although factors such as weather could potentially contribute to these patterns, weather data was not included in this analysis, so the cause of the seasonal variation cannot be determined from the trip data alone.

Casual ridership shows greater monthly variation than member ridership



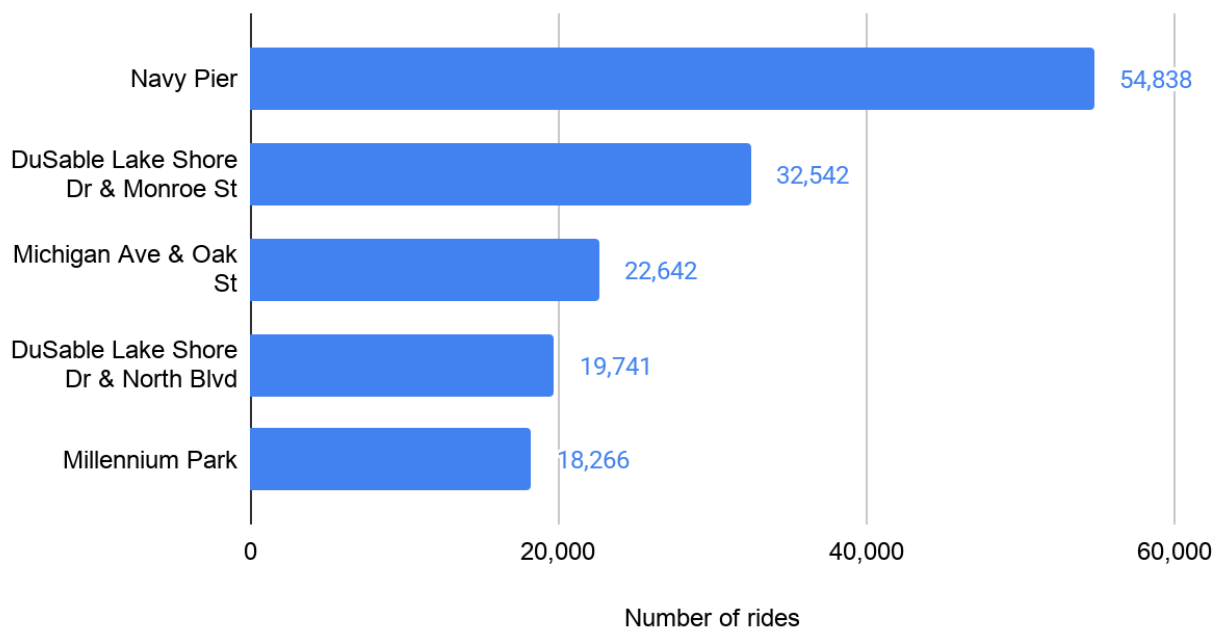
5.5 Popular Start Stations Differ Between Rider Groups

The most frequently used start stations differed substantially between casual riders and annual members. For casual riders, **Navy Pier** was the highest-volume start station with **54,838 rides**, followed by locations including DuSable Lake Shore Dr & Monroe St, Michigan Ave & Oak St, DuSable Lake Shore Dr & North Blvd, and Millennium Park.

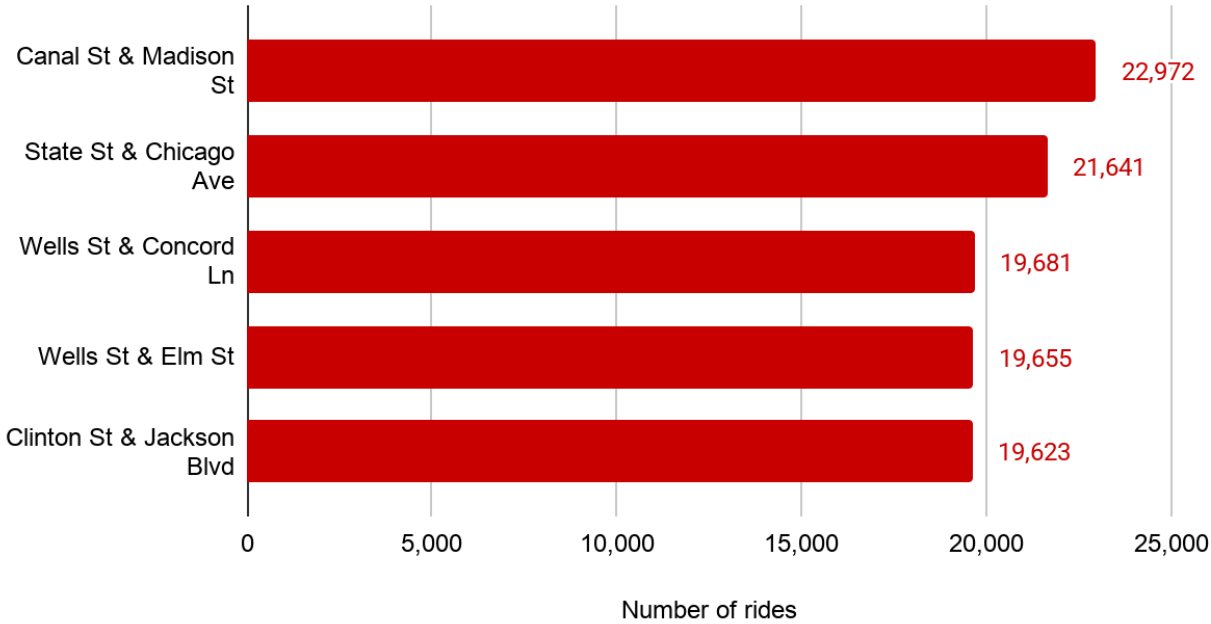
For annual members, **Canal St & Madison St** was the highest-volume start station with **22,972 rides**, followed by State St & Chicago Ave, Wells St & Concord Ln, Wells St & Elm St, and Clinton St & Jackson Blvd.

These results indicate that the two rider groups have distinct location patterns. Because some rides contained missing station information, this analysis includes only rides with recorded start-station names. The station data also does not establish whether riders are tourists, commuters, residents, or using the bikes for another purpose.

Top 5 start stations for casual riders



Top 5 start stations for members



6. Recommendations

Based on the differences identified between casual riders and annual members, Cyclistic should consider three targeted strategies to encourage casual riders to convert to annual memberships.

1. Time membership promotions around periods of high casual ridership.

Casual usage increases toward the weekend and shows greater seasonal variation than member usage. Cyclistic could test membership offers or promotional campaigns during weekends and higher-ridership months, when casual riders are more actively using the service.

2. Target high-volume casual start stations.

Casual riders show distinct location patterns, with particularly high ride volumes at stations such as Navy Pier and Millennium Park. Cyclistic could prioritize membership marketing at high-volume casual start stations, especially during periods of strong casual demand.

3. Tailor membership messaging toward casual riders taking longer-duration rides.

Casual riders have a higher median ride duration than annual members. Cyclistic could test messaging that communicates the potential value of annual membership to casual riders taking longer-duration trips. Any claims about financial savings should first be validated against Cyclistic's actual membership and casual-use pricing.

These strategies should initially be tested through targeted campaigns and evaluated using conversion rates and other relevant marketing metrics before being implemented more broadly.

7. Limitations

This analysis has several limitations that should be considered when interpreting the results.

- **Individual riders cannot be identified.** The dataset contains trip-level records rather than unique customer identifiers, so the analysis cannot determine how frequently a specific person rides or compare the number of individual casual customers with individual members.
- **Trip purpose is unknown.** Patterns such as weekday morning and afternoon peaks may be consistent with routine travel, but the data does not establish whether rides were for commuting, recreation, tourism, or another purpose.
- **Station data is incomplete.** Some rides have missing start or end station information. Station-specific analyses therefore represent only rides with recorded station names.
- **External factors were not analyzed.** Weather, holidays, events, pricing, and promotions could influence ridership patterns but were outside the available trip dataset.

- **Membership economics were not evaluated.** Without Cyclistic's actual pricing and customer-level purchasing information, the analysis cannot determine exactly when annual membership becomes financially preferable for a casual rider.

8. Conclusion

This analysis demonstrates that annual members and casual riders use Cyclistic bikes in meaningfully different ways. Members generate the majority of rides and show more regular weekday and time-of-day patterns, while casual riders typically take longer rides and display stronger weekend, seasonal, and location-based patterns.

These differences provide Cyclistic with several opportunities to make membership marketing more targeted. Rather than applying the same strategy to all casual riders, Cyclistic can test campaigns during periods and at locations where casual usage is highest and tailor messaging around the characteristics observed in casual-rider behavior.

Future analysis incorporating customer-level behavior, pricing, weather, promotions, and campaign conversion data could help determine which strategies are most effective at converting casual riders into annual members.